JULIAN PIETRAROIA

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EDUCATION

University of British Columbia – GPA: 3.9 *Expected April 2027*
Bachelor of Applied Science in Manufacturing Engineering
Dean's Honour List
Top 10% of Faculty of Applied Science
Relevant Coursework: Industrial Automation, Manufacturing Processes, Mechanics of Materials, Mechanical Vibrations, Engineering Measurements & Instrumentation, Machine Dynamics, Software Design for Engineers

EXPERIENCE

Hardware Engineer – Stealth Startup ($85M raised), San Francisco, CA *June 2026 – Sep 2026*
- Engineered and manufactured a 1-DOF UMI (Universal Manipulation Interface) gripper now used by 200+ data collectors to scalably capture robot manipulation demonstrations
- Developed a full 22-DOF articulating glove that individually tracks every finger joint for high-fidelity hand-pose and teleoperation data
- Deployed a custom 1- and 2-DOF actuated gripper across teleoperated data collection stations to gather 1,000 hours of manipulation data
- Designed a 6-DOF leader arm for scaled teleoperation stations, extending data collection from UMI-only demos to full-arm pose and trajectory capture

Manufacturing Engineering Intern – cellcentric (Toyota, Volvo & Daimler Truck), Burnaby, BC *May 2024 – Aug 2025*
- Applied Lean Manufacturing to optimize a state-of-the-art fuel cell production process, reducing cost and improving cycle times by 40%
- Led Value Stream Mapping to identify bottlenecks and calculate process capacities, driving continuous-improvement initiatives
- Performed data analysis and root cause analysis on production data to surface trends and predict failure points
- Built company-wide Power BI dashboards for production and budget reporting, improving efficiency and reducing reporting time

Mechanical Lead – Ten Ton Robotics (VexU Competitive Robotics), West Vancouver *Aug 2023 – Present*
- Won the 2025 VEX U World Championship and back-to-back Skills World Championships against 300+ university teams, leading the mechanical design of both competition robots
- As Mechanical Lead, guided a 14-member team through the full design cycle, from concept and CAD to fabrication and iteration
- Applied both subtractive (CNC) and additive (3D printing) manufacturing to produce unique, high-performance components
- Conducted FEA in Ansys on custom parts to optimize structural integrity and performance

PROJECTS

Autonomous Ping Pong Turret – Personal Project *Aug 2026 – Present*
- Engineering a turret that autonomously shoots ping pong balls into cups from 1–5 m away, tracking each cup with computer vision and solving launch angle and power per shot
- Improved shot-to-shot repeatability by designing a ratchet-driven puncher on a linear slide that cocks and fires from a single motor reversal, cutting the mechanism to one actuator
- Delivered closed-loop aiming by driving pan and tilt with Dynamixel servos over a U2D2 bus, holding the turret locked on the tracked target between shots

Line Following Robot *Apr 2026 – May 2026*
- Designed and manufactured a fully autonomous line-following robot with obstacle avoidance that generates a heat map of its environment
- Implemented sensor fusion of encoder and IMU odometry to track robot position and orientation during autonomous navigation
- Integrated IR-based line detection and proximity sensing for reliable autonomy across varied environments

Combat Robots – Personal Project *Apr 2026 – Present*
- Designing a 1 lb antweight combat robot around a brushless hub-motor spinner weapon and a 4S brushless drivetrain, sizing motors and gearing for wood-floor traction and weapon spin-up
- Engineered chassis and armor in Fusion 360, selecting carbon-fiber nylon, TPU, titanium, and AR-500 steel to survive repeated high-energy impacts
- Experimenting with high-speed ducted fans that pull the chassis into the floor, tuning fan thrust and ground clearance to convert airflow into drive traction and pushing power

SKILLS

Mechanical: Robotics, Mechatronics, GD&T, Design for Manufacturing (DFM)
Manufacturing: Injection Molding, Thermoforming, CAM (Computer-Aided Manufacturing), SPC
Design / CAD: SolidWorks, Onshape, Fusion 360, Siemens NX, Ansys (FEA)
Electrical / Controls: PLC, Microcontrollers, Firmware / Embedded Systems, Control Systems (PID), Soldering, Oscilloscope
Software: Python, C / C# / C++, MATLAB, Git, Excel, Minitab, Ladder Logic, LaTeX

AWARDS

VEX Robotics World Champions (2025) • VEX Robotics Excellence Award (2025) • VEX Robotics Skills Challenge World Champions (2024, 2025) • A.E. Coleman Leadership Award (2022) • Schulich Leader Nominee (2022)